

## BYCATCH OF *CARETTA CARETTA* IN THE NORTHERN ADRIATIC SEA

*Caretta caretta*, known as loggerhead turtle, is the most abundant sea turtle species in the Mediterranean Sea. It is a migratory species, is extremely mobile and the northern Adriatic Sea has been identified as an important foraging ground for *C. caretta* during all life stages (Casale *et al.*, 2012).

*C. caretta* has two main ecological life-stages: a pelagic stage, in which juveniles feed on pelagic preys, spending most of their time at less than 10 m depth and a neritic stage in which sub-adults and adult specimens live mainly near the seabed, feeding on benthic preys and usually not exceeding 50 m depth (Lucchetti *et al.*, 2017).

The northern Adriatic Sea, and particularly the FAO GSA-17, is considered an essential foraging ground for *C. caretta*. It is a continental shelf zone, with average of 35 m depth, in which turtles are very abundant thanks to the easy access to a lot of food, this is, indeed, a highly productive area thanks to the discharges of the Po River. Po River represents the main source of nutrients in the whole basin, sustaining a high biodiversity, important for the *C. caretta* trophic interactions (Marisaldi *et al.*, 2023).

The Mediterranean *C. caretta* population has declined in recent decades, leading the IUCN to list it as an “endangered species” (Lucchetti *et al.*, 2016). In the northern Adriatic there are several Natura 2000 sites (SIC, SPA and both SIC and SPA) and most of them have among their protection target *C. caretta*, according to the “Habitat Directive” (Fortuna *et al.*, 2018). We can see the *C. caretta* and Natura 2000 sites distribution within the FAO GSA-17 in the **MAP 1**.

The bathymetric and productivity characteristics of the northern Adriatic Sea make it also an ideal fishing ground with a high fishing vessel density within this area which is one of the most overfished in the Mediterranean Sea (Lucchetti *et al.*, 2016), moreover here there are some of the most important Italian harbours (Pulcinella *et al.*, 2019). These human uses can affect *C. caretta* and they are displayed inside the **MAP 2**.

Bycatch is the major threat to *C. caretta* survival (Lucchetti *et al.*, 2016), within the Mediterranean Sea, every year, about 150.000 of them are involved in bycatch events with more than 50.000 deaths (Casale *et al.*, 2010). *C. caretta* bycatch occurs mainly in the northern Adriatic Sea (Pulcinella *et al.*, 2019) with more than 10.000 accidental catches taking place here annually (Lucchetti *et al.*, 2016). In the FAO GSA-17 we have a consistent overlapping between fishing activities and *C. caretta* habitats (Lucchetti *et al.*, 2017) as visible within the **MAP 3**: considering a buffer of 20 km around *C. caretta* sightings [it is estimated that their maximum average speed of movement is 20 km/day (Casale *et al.*, 2012)] and a buffer of 5 km around the fishing vessel density with values higher than 5 h per km<sup>2</sup>, we have an important overlapping between them, explaining the high bycatch rate of this area. The main bycatch hotspot was identified in the area between Ancona's harbour and the Po River delta (Pulcinella *et al.*, 2019), with the estimate annual catch for fishing vessel near the Po River delta that is about 5,30 (Lucchetti *et al.*, 2017). Below 43,5° N of latitude we have lower bycatch values (Pulcinella *et al.*, 2019).

*C. caretta* bycatch in the northern Adriatic Sea occurs mainly by longlines, trawls and fixed nets with the mortality that can be both direct and delayed by drowning, release in comatose state or with netting pieces entangled on their bodies (Lucchetti *et al.*, 2017).

Possible solutions to reduce the bycatch problem could be: upgrading knowledge about the distribution and the migration patterns of *C. caretta*, installation of Turtle Excluder Devices (TEDs) to nets and limiting the overall fishing effort in the areas in which the species' abundance is higher. The mortality rate could also be reduced by limiting the haul duration and allowing comatose turtles to recover on the vessels before their release (Casale *et al.*, 2004). According to the overlap between species' abundance and fishing vessel density inside the Natura 2000 sites (**MAP 3**) it could be useful to add more restrictions inside these areas.

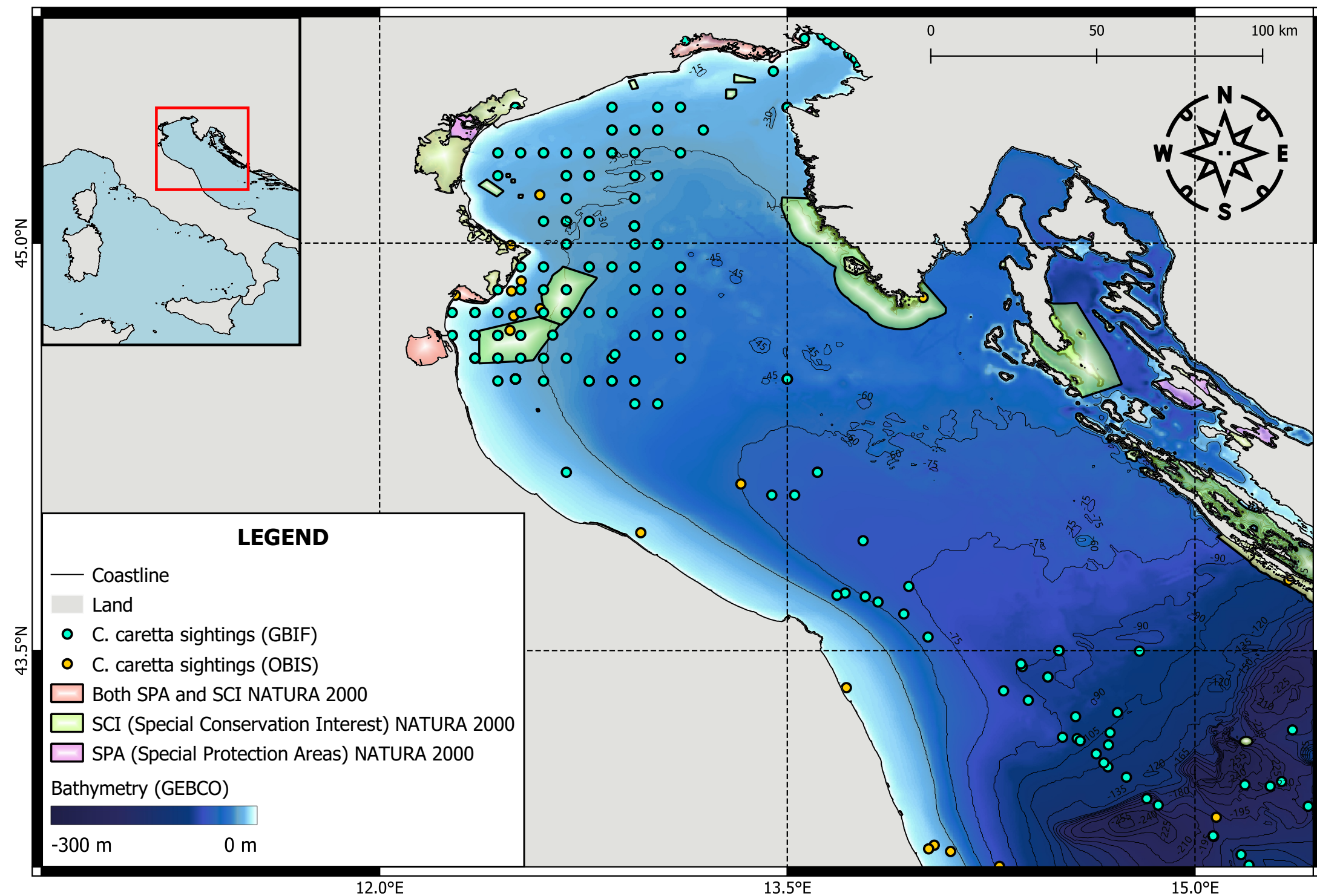
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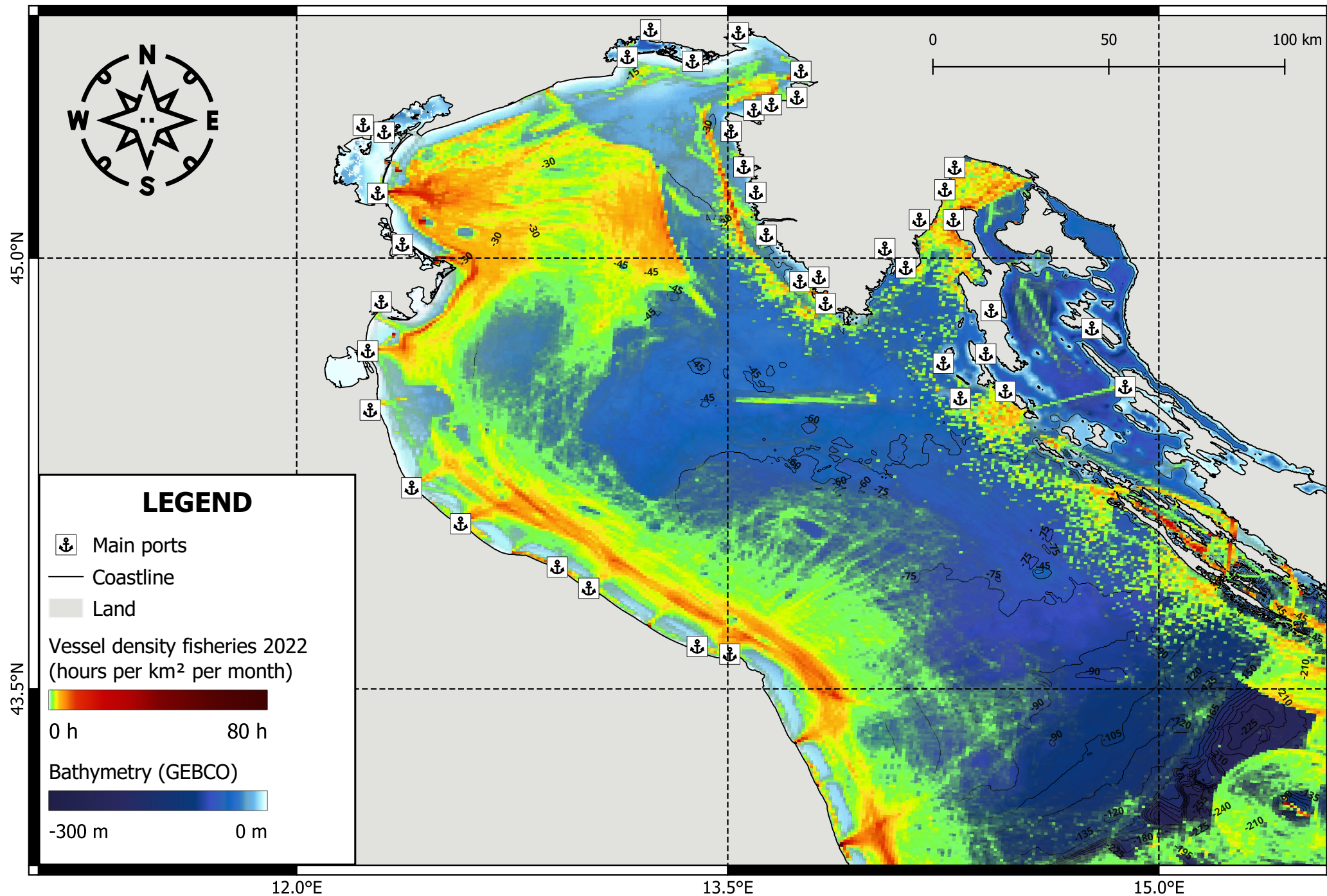
## DATA SOURCES:

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# CARETTA CARETTA DISTRIBUTION AND PROTECTED AREAS



# MAIN PORTS AND FISHING VESSEL DENSITY



# OVERLAPPING BETWEEN CARETTA CARETTA DISTRIBUTION AND HUMAN USES

